

Foundations of Artificial Intelligence

Part I: Introduction & Intelligent Agents

Department of Computer Science

Lecture Roadmap

1. Chapter 1: Defining the Field
2. Chapter 2: The Agent Framework

Learning Objectives

By the end of this lecture, you should be able to:

- Distinguish the four classic definitions of AI.
- Explain why the **rational agent** view is central in modern AI.
- Specify a task using **PEAS** with concrete metrics.
- Classify environments (observable, Random, sequential, etc.).
- Choose an appropriate agent structure for a given problem.

Chapter 1: Defining the Field

What is Artificial Intelligence?

AI is historically viewed through four distinct lenses:

Human-Centric

- **Thinking Humanly:** Cognitive modeling.
- **Acting Humanly:** The Turing Test.

Rationality-Centric

- **Thinking Rationally:** Laws of thought (Logic).
- **Acting Rationally:** The Rational Agent.

This course focuses on the Rational Agent approach: acting to achieve the best outcome.

Four Views of AI: Strengths and Limits

View	Main Idea	Typical Limitation
Thinking Humanly	Match human cognitive processes	Hard to validate internal thought processes
Acting Humanly	Indistinguishable behavior from humans	Human-like behavior is not always optimal
Thinking Rationally	Derive correct conclusions via logic	Real-world knowledge is incomplete/uncertain
Acting Rationally	Select actions that maximize expected performance	Requires modeling uncertainty and trade-offs

A Brief Historical Arc of AI

- **1950s–1960s:** Symbolic AI and early optimism.
- **1970s–1980s:** Knowledge-based systems and expert systems.
- **1990s–2000s:** Probabilistic reasoning and machine learning become central.
- **2010s–Today:** Deep learning, foundation models, and hybrid systems.

Book takeaway: progress accelerates when representation, learning, and computation improve together.

The Foundations of AI

AI is a "synthetic" discipline, drawing from:

- **Mathematics:** Probability (uncertainty) and Logic (certainty).
- **Economics:** Decision theory and utility maximization.
- **Neuroscience:** Hardware for intelligence (biological).
- **Control Theory:** Designing systems that maximize an objective function over time.

Reference Reading for This Part

- **Russell & Norvig (4th ed.):** Chapter 1 (Introduction), Chapter 2 (Intelligent Agents).
- **Poole & Mackworth (3rd ed.):** Chapters on agents, environments, and problem formulation.

Suggested strategy:

1. Read definitions first (AI views, rationality, PEAS).
2. Then map each definition to one practical example.
3. Finally, compare which assumptions break in real-world settings.

Chapter 2: The Agent Framework

The Agent-Environment Interaction

An **Agent** is anything that:

1. Perceives its **Environment** through **Sensors**.
2. Acts upon that environment through **Actuators**.

Percepts $\rightarrow$ [Agent Function] $\rightarrow$ Actions

Agent Program: The concrete implementation of the mathematical Agent Function.

Formal Agent Definition

Let:

- $\mathcal{P}$ be the set of possible percepts,
- $\mathcal{A}$ be the set of possible actions.

Then an agent function is:

$$f: \mathcal{P}^* \rightarrow \mathcal{A}$$

Interpretation:

- Input: entire percept history.
- Output: next action.

Designing for Success: PEAS

Before writing code, we must specify the **PEAS** for the task:

Performance The metric for success (e.g., safety, speed, profit).

Environment The "world" (e.g., city streets, a chessboard).

Actuators The tools for action (e.g., steering wheel, display).

Sensors The inputs (e.g., cameras, keyboard, GPS).

PEAS Example: Autonomous Taxi

Component	Example Specification
Performance	Safety, legality, trip time, passenger comfort, energy efficiency
Environment	Urban/suburban roads, pedestrians, weather, traffic rules, other drivers
Actuators	Steering, throttle, brake, indicators, display, audio prompts
Sensors	Cameras, LiDAR, radar, GPS, IMU, wheel encoders, microphone

PEAS Example: Medical Diagnosis Assistant

Component	Example Specification
Performance	Diagnostic accuracy, calibration, turnaround time, patient safety
Environment	Hospital workflow, EHR systems, lab pipelines, changing clinical guidelines
Actuators	Ranked differential list, recommendations, alerts, report generation
Sensors	Symptoms, vitals, labs, imaging reports, history, clinician feedback

Task Environment Properties

The difficulty of an AI problem is defined by its environment:

- **Observable:** Fully vs. Partially (Can we see everything?).
- **Deterministic:** vs. Random (Is the next state predictable?).
- **Episodic:** vs. Sequential (Do current actions affect the future?).
- **Static:** vs. Dynamic (Does the world change while we think?).
- **Discrete:** vs. Continuous (Chess vs. Driving).

Environment Classification by Example

Task	Observability	Dynamics	Outcome Model	State Type
Chess	Fully observable	Static (turn-based)	Mostly deterministic	Discrete
Poker	Partially observable	Static (turn-based)	Random (hidden cards)	Discrete
Robot vacuum	Partially observable	Dynamic	Random	Hybrid
Highway driving	Partially observable	Dynamic	Random	Continuous

The Hierarchy of Agent Structures

Modern AI agents are categorized by their complexity:

- **Simple Reflex:** Action based only on current percept.
- **Model-Based:** Maintains internal state to track the "unseen."
- **Goal-Based:** Acts to reach a specific destination or state.
- **Utility-Based:** Acts to maximize "happiness" (useful for trade-offs).
- **Learning Agents:** Can improve their performance over time.

When to Choose Each Agent Type

- **Simple Reflex:** Use when environments are fully observable and rules are stable.
- **Model-Based:** Use when some state is hidden and must be inferred.
- **Goal-Based:** Use when planning to a target state is required.
- **Utility-Based:** Use when multiple goals conflict and trade-offs matter.
- **Learning Agent:** Use when environment is complex, non-stationary, or hard-coded rules fail.

Learning Agent Architecture (Conceptual)

A standard learning agent can be decomposed into:

- **Performance Element:** Chooses external actions.
- **Learning Element:** Improves the performance element from experience.
- **Critic:** Evaluates behavior against the performance measure.
- **Problem Generator:** Proposes exploratory actions to gather informative experience.

Key Takeaways

1. Rationality is about **expected performance**, not perfect outcomes.
2. PEAS is the first engineering step before implementation.
3. Environment properties determine algorithmic difficulty.
4. Agent architecture should match uncertainty, goals, and learning needs.

Questions?